6CCSANLP – Coursework

Large Language Model based Open Question Answering

Submission deadline: Thursday, Dec 17, 2026

Introduction to Open Question Answering

Open Question Answering (Open QA) is a subfield of Natural Language Processing (NLP) that focuses on enabling systems to answer questions using large, unstructured collections of text, such as web documents or Wikipedia articles. Unlike closed-domain QA, which relies on predefined knowledge bases or specific datasets, Open QA models must retrieve, comprehend, and synthesize information from diverse sources in real time. This makes Open QA a challenging yet powerful task that lies at the intersection of information retrieval, reading comprehension, and reasoning. Its applications span intelligent search systems, digital assistants, and automated knowledge discovery, reflecting the broader goal of enabling machines to understand and reason over human language at scale.

For this coursework, we employ the HotpotQA dataset, a large-scale benchmark designed to test multi-hop question answering over Wikipedia articles. HotpotQA requires models not only to find relevant information, but also to **reason across multiple supporting documents** to arrive at a correct answer. The dataset is publicly available and well-documented: you can view its official homepage here and consult the detailed README, which describes how to install, evaluate, and use the dataset, here.

Paragraph A: Peter Marc Jacobson
Peter Marc Jacobson (born October 27, 1957) is an American television writer, director and producer, and actor. He is best known as the co-creator of the popular sitcom "The Nanny", which he created and wrote with his then wife actress Fran Drescher, who was the star of the series. He was often credited as Peter Marc in his early acting roles.

Paragraph B: Fran Drescher
Francine Joy "Fran" Drescher (born September 30, 1957) is an American actress and activist. She is best known for her role as Fran Fine in the hit TV series "The Nanny" (1993–99), and for her nasal voice and thick New York accent.

```
Q: Peter Marc Jacobson is best known as the co-creator of the popular sitcom
   "The Nanny", which he created and wrote with his then wife an actress born
   in which year ?
A: 1957
```

Fig. 1. An example from the HotpotQA dev set. Answering the question requires retrieving relevant information from multiple documents. Sentences highlighted in green are supporting facts for this question.

Methodological Approaches

To explore the Open QA task, students can select from several methodological approaches introduced throughout the course. The first option is the **prompting-based method** (Week 7), which leverages large

language models (LLMs) through carefully designed prompts to elicit reasoning and answers without additional fine-tuning.

The second approach is **Supervised Fine-Tuning (SFT)** (Week 9), where models are trained on labelled data. In this step, students are free to select datasets beyond HotpotQA's training set to enhance performance or adapt to specific domains.

A third option is the **Retrieval-Augmented Generation (RAG)** method (Week 10), which combines retrieval mechanisms with generative models. For practical reasons, students are encouraged to use the **Wikipedia corpus provided by HotpotQA**, as constructing a large-scale retrieval system from scratch would be time-consuming.

Finally, the **LLM reasoning-based method** (Week 11) focuses on structured reasoning and chain-of-thought generation using advanced LLMs. Students may also experiment with hybrid systems that combine these techniques.

For the coursework implementation, you can use Mistral APIs. You can register for a Mistral account at https://auth.mistral.ai/ui/login. To support implementation and experimentation using Mistral APIs, a Colab demo is provided here, offering example workflows and evaluation scripts for the methods mentioned above.


```
"answer":
{
  "5ae66d515542992ae0d16346":"1927",
  "5a83aaeb5542996488c2e483":"Graduation",
  "5a8c66db554299585d9e3683":"2007",
  "5ae16ea85542990adbacf790":"31 July 1975",
  "5ac10832554299012d1db687":"several",
  "5ae2c4b8554299495565db31":"New York City",
  "5a78d38455429970f5fffd94":"Julian Assange",
  "5adce4f65542992c1e3a2474":"United States Department of Health and Human Services",
  "5ab6654455429954757d3281":"genus of flowering",
  "5adf39ce5542995534e8c76d":"Shakespeare performances throughout India",
  "5a7619765542992db9473728":"Randers, Denmark",
  "5a77a3905542992a6e59df40":"Duke Blue Devils",
  "5ac38ce255429939154137c2":"water poured over ice",
  "5a8a38135542996c9b8d5e59":"Chancellor of Germany",
  "5ac3bd0c554299204fd21e21":"filmmaker",
  "5adcaa7f5542990d50227cdb":"graduate students",
  "5a85b0c85542994c784ddb40":"Texas Tech University",
   "5add6b595542992200553b0c":"Atlanta, Georgia",
 }
```


Fig. 2. An example of a generated answer that is accepted by the evaluation script. The generated answer is aligned with its corresponding question ID.

Evaluation Setup and Guidelines

Since the HotpotQA dataset does not provide public annotations for the official test set, this coursework adopts a modified evaluation procedure. Students are instructed to use the **development set with the full Wikipedia corpus** as the primary testing set. For methods that require model development or validation (such as fine-tuning or hyperparameter tuning), students should **split a small subset from the training data** for this purpose.

If the chosen approach is **prompting-based**, **RAG-based**, or involves **sampling-based reasoning**, evaluation solely on the **development set** will be sufficient. Students should use the official evaluation script (available here) to assess their models' performance, ensuring consistent and comparable results across submissions.

Marking Criteria and Report Requirements

The coursework will **not be graded based on model performance**, but rather on the overall **research quality, originality, and clarity of presentation**. Assessment will be conducted according to the following three aspects:

1. **Novelty** – Evaluate whether your proposed demonstrates creativity inspired from the techniques introduced in this module. Students are encouraged to extend, modify, or combine methods (e.g., prompting, SFT, RAG, or reasoning-based approaches) to design unique solutions to the Open QA task.
2. **Organisation and Experimental Support** – Assess how effectively your experimental design supports your proposed approach. Strong reports should include clear experimental setups, performance analyses, and evidence such as **ablation studies** and **case studies** to substantiate your claims.
3. **Report Writing** – The written report must follow an academic paper format and include the following sections: **Introduction/Literature Review/Methodology/Experiment/Conclusion. References** section should be appended, containing **at least 10 references**.

Students are advised to consult our **previous work on this dataset** available on arXiv here for reference in writing style and structure. The report should follow the **ACL 2026 LaTeX template**, which can be accessed here.

The **total length** of the report (excluding references) should be **between 4 and 6 pages**. Submissions exceeding or falling short of this range will receive a penalty.

Additional Requirements

Academic Integrity – Any form of plagiarism or academic misconduct will result in a mark of **zero (0)** for the coursework. Students must comply with the Academic Misconduct Policy at King's College London.

Reproducibility – All submitted code must be **fully runnable**, and the **reported results must match the outputs produced by the code**. Discrepancies between the report and the executable results may affect the final grade.

Using the Mistral API - This module is supported by Mistral. You can register for a Mistral account at https://auth.mistral.ai/ui/login. Please also provide your team information at KEATs; a capped budget will be allocated to each student.

Use of Generative AI: This module adopts "**Model 2: Authorised use of Generative AI tools to perform literature search**" from the department's AI policy in the Informatics Handbook. Generative AI tools can be used for spelling/grammar correction and help gather relevant information such as literature review. For coding, it can be used for refactoring of routines and small-scale bug fixing. **An acknowledgement is required in the form of a brief statement identifying the extent to which the LLM tool was used along with the prompt.**

Submission format

The submission for this assessment should include:

1. **A group report** that meets the above requirements.
2. **A readme file** outlining the Python environment setup and instruction for running the code.
3. **Python code**.
4. **Predictions on the development (Dev) set,** formatted according to the Dev example provided on the HotPotQA webpage.